SDG 8 · DECENT WORK AND ECONOMIC GROWTH · PROBLEM 8-A

RAKSHA

Heat and air-safety monitor for informal workers. Workstation node for brick kilns, welding shops, tailoring units, and construction sites.

Simulate at	https://logicraftstudios.tech/editor/
Project file	simulation/raksha-logicraft.zip
Board	ESP32 DevKit V1
Round	3 hours · pitch in seven sections · 3-minute defence

1. What the node does

RAKSHA watches heat, process-air risk, and falls. It warns the worker with a vibration motor and an RGB lamp, because a buzzer is lost in kiln and welding noise. A fan relay and a beacon relay act on the room. Exposure time is stored against the worker who checked in with an RFID card, and it counts only while that worker is in the zone.

Three decisions carry the design:

- **Presence gate.** PIR or IR must see a person before heat or air time accumulates. An empty bench does not invent hours.
- **Free-fall gate.** A fall is a short free-fall, then an impact, then a still posture. Shaking the bench never produces the free-fall, so kiln vibration is rejected.
- **Relays fail off.** Both relays are off at boot and off on a sensor fault. Past a real threshold the fan closes. On a latched fall the beacon closes.

> This kit has no gas sensor. The air result is a process-risk score: sustained shop noise, humidity that is not clearing, and heat, gated by presence. Say that in the pitch. The KY-037 is not a ppm meter.

2. Pipeline

```
POWER → BOOT → SENSE → ESTIMATE → DECIDE → ACT → RECORD → SERVE
```

Stage	When	Result
Power	Once, before logic	12 V through the HW-411, measured at 5.0 V, feeds the ESP32 VIN, the relay board, and the motor supply.
Boot	Every reset	Relay pins driven inactive first. Then I2C, DHT22, and RFID are checked. A failed critical sensor stays in fault. Relays stay off.
Sense	10 ms and 2 s	MPU6050 every 10 ms. DHT22, NTC, barometer, sound, potentiometer, PIR, and IR every 2 s. RFID on tap.
Estimate	Same clocks	Heat index, three-thermometer vote, 0–100 process-risk score, presence, acceleration magnitude, tilt.
Decide	Same clocks	Shift, heat band, air alert, fall state, fault. Commands need the PIN.
Act	On change	OLED, RGB, vibration, fan relay, beacon relay.
Record	15 s and every edge	One CSV row in flash: worker, heat, risk, presence, fall, relays.
Serve	Continuous	Dashboard on the RAKSHA access point at 192.168.4.1 if the workshop router is down. Ack, fan, and threshold need the PIN.

The 10 ms fall path never waits on the DHT22, the filesystem, or HTTP. Every threshold and pin lives in one `config.h`.

3. Simulation on Logicraft Studios

The simulation runs in the browser at **https://logicraftstudios.tech/editor/**. It is the Velxio editor. It compiles the ESP32 sketch and runs it against the wired canvas. The project is a Wokwi zip, which that editor opens directly.

Open the project

1. Open https://logicraftstudios.tech/editor/ .
2. If a “Start a new project” panel is showing, press Cancel.
3. Press the open-folder control, *Open project (.vlx Logicraft Studios or .zip Wokwi)*.
4. Choose `simulation/raksha-logicraft.zip`.
5. Press Libraries and confirm these four are installed: DHT sensor library for ESPx, Adafruit GFX Library, Adafruit SSD1306, Adafruit BusIO. Install any that are missing.
6. Press the compile / run control. Open Serial at 115200.
7. In the serial send box, type `RUN` and send it.

The scripted run lasts about 20 seconds. It uses the same heat-index function, the same risk weights, and the same fall state machine as the bench firmware. The serial monitor is the simulation record for the pitch.

> **Canvas stand-ins.** Logicraft has no RC522 and no KY-037. The green CARD button is the RFID tap. The left potentiometer is the sound envelope. The buzzer is the vibration motor on GPIO 16. The barometer, the IR sensor, the buck converter, and the 74HC14 stay on the bench; they are not on this canvas. The heat vote in the simulator is DHT22 against the NTC only.

What is on the canvas

Canvas part	GPIO	Bench part it represents
ESP32 DevKit V1	—	ESP32
SSD1306 OLED	21 SDA, 22 SCL	OLED. The I2C LCD stays in the bag.
MPU6050, +Z up	21, 22	MPU6050
DHT22, slider for °C and RH	4	DHT22
NTC module, β = 3950	36 (VP)	NTC divider
PIR, “Simulate Motion”	34	PIR after the 74HC14
Potentiometer, start at 15%	32	KY-037 analog
Potentiometer, start at 45%	33	Threshold potentiometer. 45% maps to threshold 57.
Green CARD button	27, pull-up	RFID. Press once to open the shift. Press again during an alarm to acknowledge.
Relay module	17 (TX2), active-low	Fan relay
Relay module	26, active-low	Beacon relay
Red, green, blue LEDs, 220 Ω	25, 13, 14	RGB LED
Buzzer	16 (RX2)	Vibration motor through the 2N2222A and the 1N4007

Scripted scenario

Type `RUN`. The firmware holds the threshold at 57 and plays eight steps. Reference values below come from the same formulas as the sketch. The serial line is the result to screenshot.

Time	Step	What you must see
0.5 s	Card, then 30 °C / 55% RH, worker present, sound envelope 0.10	`SHIFT worker=SIM-01`. Heat index about 31.9 °C, band CAUTION, risk about 13, fan off, beacon off.
2.5 s	Clap: 28 °C / 70% RH, envelope 0.12	Heat index about 30.7 °C, still CAUTION, risk about 20, air off, fan off. A short noise does not trip the fan.
5 s	Settled noise: same climate, envelope 0.90	Risk about 59, `air=1`, `fan=1`. The band is still CAUTION, so the fan closed because of process risk, not because the heat band reached danger.
8 s	Shake for 2.5 s, magnitude stays above 1 g	No `FALL FREEFALL` line. Fall stays IDLE.
10.5 s	Drop: 100 ms at 0.15 g, then 3.2 g, then a tilted still posture and no presence	`FALL FREEFALL`, `IMPACT`, `CONFIRM`, `ALARM`. Beacon relay on. Buzzer on. Fan releases, because the air alert requires a person present.
15 s	Second card tap	`ACK`. Fall returns to IDLE. Beacon off.
17 s	DHT removed	`FAULT DHT relays=OFF`. Red LED blinks. Both relays off.
19 s	End	`RUN done`.

RUN applies the settled 10-second sound envelope so the pitch fits in one pass. In live mode the envelope really is a 10-second average: a clap dies away, and the sound potentiometer has to be held up. Live mode also reads the DHT22 slider, the NTC slider, the PIR "Simulate Motion" button, both potentiometers, and the CARD button. Other serial commands: CARD, DROP, SHAKE, UNPLUG, HELP.

Screenshot for the Simulation slide

Capture the serial transcript from `RUN start` through `FAULT DHT relays=OFF`, with the canvas visible enough to show the fan relay during the noise step and the beacon relay during the alarm. That single capture is the simulation section of the pitch.

4. Algorithms

Heat index

Convert DHT temperature to °F. If the simple screening value averages under 80 °F with the air temperature, use that screening value. Otherwise use the Rothfusz regression, including the low-humidity and high-humidity corrections, then convert back to °C.

Heat index °C	Band	While a worker is present
below 27	None	Green. Counters idle.
27 to 32	Caution	Amber. One short vibration pulse. Exposure seconds increment.
32 to 39	Extreme caution	Amber. Exposure seconds increment.
39 to 51	Danger	Red. Repeating vibration. Fan on. Danger seconds increment.
above 51	Extreme	Same as danger.

Temperature vote

Compare the DHT22 with the NTC and with the barometer temperature. Heat is invalid only when the DHT misses both of the others by more than 8 °C. In the simulator there is no barometer, so the comparison is DHT against NTC. Invalid heat forces the relays off and the OLED shows a sensor fault. A single disagreeing sensor does not do that.

Process-risk score

```
stagnation   = clamp((RH − 65) / 20, 0, 1)
heat_couple  = clamp((heat_index_C − 27) / 12, 0, 1)
risk         = 100 × (0.50 × sound_ema + 0.30 × stagnation + 0.20 × heat_couple)
threshold    = 30 + (potentiometer × 60)     range 30 to 90
air_alert    = worker present AND heat valid AND risk ≥ threshold
```

`sound_ema` is a 10-second exponential average of the KY-037 analog pin, normalised to 0–1. The weights are 0.50, 0.30, and 0.20.

Fall machine

Magnitude is the root-sum-square of the three accelerometer axes in g. Mount the board with +Z up. Tilt is computed only while the magnitude is between 0.7 g and 1.3 g, so the impact spike itself cannot look like a posture change.

From	To	Rule
Idle	Free-fall	Magnitude stays under 0.4 g for at least 80 ms.
Free-fall	Impact	Magnitude exceeds 2.5 g within 500 ms. Otherwise return to idle.
Impact	Confirm	Tilt passes 60° from upright and the PIR is quiet.
Confirm	Alarm	That posture holds for 2 s. If the worker moves, return to idle.
Alarm	Idle	The same worker's card is tapped again, or a fault clears the node.

The alarm latches the beacon and the vibration. Gyro data is not used.

5. Bench hardware

Set the HW-411 to 5.0 V with a multimeter before the ESP32 is connected. That rail feeds VIN and the relay board. The vibration motor is powered from 5 V through the 2N2222A, with the 1N4007 across the motor, cathode at 5 V. The ESP32 3.3 V pin does not feed the motor or the relays.

Power the 74HC14 from 3.3 V and use it to clean the PIR and the KY-037 digital pin. A 5 V high into an ESP32 GPIO damages the pin. Analog signals stay on ADC1, because Wi-Fi uses ADC2.

Function	GPIO	Note
I2C SDA / SCL	21 / 22	OLED 0x3C, MPU6050 0x68, barometer 0x76 or 0x77
RFID RC522	18, 19, 23, SS 5, RST 27	SPI. Second tap acknowledges. Unknown card is refused.
DHT22	4	10 kΩ to 3.3 V
KY-037 analog	32	ADC1
Potentiometer	33	ADC1, threshold
PIR via 74HC14	34	Input only
IR proximity	35	Input only
NTC divider	36	10 kΩ from 3.3 V to the node, NTC to ground. β = 3950, R0 = 10 kΩ
Vibration base	16	1 kΩ into the 2N2222A
Fan relay / beacon relay	17 / 26	Active-low. Pull each input to the off level so a reset cannot glitch the load.
RGB	25, 13, 14	220 Ω, common cathode

Leave unwired: the 4×4 keypad, the I2C LCD, the spare 1-channel relay, the 2N3906, and the IRF9540. The keypad's eight pins collide with this map. RFID second-tap is the offline acknowledge.

Measurements to write on the results slide

Measurement	Pass
Buck, unloaded and with the ESP32 connected	4.8 V to 5.2 V both times
NTC at room temperature and in a warm hand	Warm hand reads higher. Record both.
Fall trial	Ten seconds of shaking stays idle. A drop onto foam from about 30 cm reaches alarm.
Reset while a relay is on	Both relays are off while the OLED is still on the self-test line.

6. Faults to rehearse

Action	Required result
Unplug the DHT22 during a shift	Fault on screen. Both relays off. The old temperature is not left up as a safe reading.
Shake the MPU for 10 s	Fall stays idle.
Drop the MPU onto foam	Alarm, vibration, beacon.
One clap, then 10 s of noise with a hand on the PIR	The clap does nothing. Sustained noise with presence turns the fan on.
Leave the zone	After 60 s the exposure counter pauses. It is not cleared.
Reset while the fan is on	The relay drops at boot and stays off until a new shift is in danger or air alert.
Unknown RFID card	Shift does not start.
Dashboard command with the wrong PIN	Rejected. State unchanged.
No router	OLED, vibration, and the CSV continue. The page loads on the RAKSHA access point.

7. Pitch order

The round asks for seven sections, then a defence of about three minutes.

1. **Introduction.** RAKSHA, a shop node for kiln, welding, tailoring, and construction work.
2. **Problem.** Heat, fumes, and falls, with no instrument and no per-worker record.
3. **Approach.** Walk the pipeline. Say the presence gate, the free-fall gate, and the relays that fail off.
4. **Simulation.** The Logicraft serial capture from `RUN`.
5. **Results.** The four bench measurements, then the one-minute hardware demo if the board is built.
6. **Edge cases.** Shake, empty room, unknown card, Wi-Fi loss, DHT removed. Offer the shake and the DHT pull to the judges.
7. **SDG 8.** The worker gets a warning they can feel in a loud shop. The employer gets minutes of exposure tied to a person, and the fan is the first corrective action.

Full stage contracts, the CSV columns, and the `config.h` names are in docs/pipeline.md. The three-hour clock is in docs/runbook.md. This PDF is the document to print and the procedure for the simulator.